```
clear all
```

```
Warning: Class 'sym' is defined in a class folder and takes precedence over a function with the
same name that is earlier on the MATLAB path. In a future release, class 'sym' will no longer be
given precedence.

Click here for the locations of the conflicting items.
Click here for guidelines to avoid this warning.
```

```
clc
```

Funwork 4

Phillipp Gery

```
syms theta1 theta2 x_pos theta1_dot theta2_dot x_dot
syms theta1_ddot theta2_ddot x_ddot
syms u1 u2 u3 x

%get Results from Funwork1 means Model
load("../Funwork3/Funwork3_Gery_Results.mat");
disp("non-linear model")
```

```
non-linear model
```

```
disp(f_u3)
```

$$
\left(\begin{array}{c}
-\frac{840\,u_1 - 2943\,\sin(2\,\theta_1) + 300\,\dot{\theta}_1^{\,2}\,\sin(\theta_1) + 135\,\dot{\theta}_2^{\,2}\,\sin(\theta_2) - 36}{} \\
\frac{3600\,u_2\cos(2\,\theta_2) - 26487\,\sin(\theta_1 - 2\,\theta_2) - 91233\,\sin(\theta_1) - 22800\,u_2 + 4725\,\dot{\theta}_2^{\,2}\,\sin(\theta_1 - \theta_2) + 750\,\dot{\theta}_1^{\,2}\,\sin(}{} \\
-\frac{2700\,\sin(\theta_1 - \theta_2)\,\dot{\theta}_1^{\,2} + 1215\,\sin(\sigma_5)\,\dot{\theta}_2^{\,2} + 13600\,u_3 - 26487\,\sigma_1 + 26487\,\sin(\theta_2}{}
\end{array}\right.
$$

where

$$\sigma_1 = \sin(2\,\theta_1 - \theta_2)$$

$$\sigma_2 = \cos(2\,\theta_1 - \theta_2)$$

$$\sigma_3 = \cos(\theta_1 - 2\,\theta_2)$$

$$\sigma_4 = 5\cos(2\,\theta_1) + 9\cos(\sigma_5) - 26$$

$$\sigma_5 = 2\,\theta_1 - 2\,\theta_2$$

Problem 1.

(4 pts) Discretize the linearized model about the equilibrium pair from FunWork 3 and use it as the design model. You as a designer select the sampling period .

```
h = 0.01;

sys_c = ss(A_u3, B_u3, C_u3, D_u3);

% continuous-to-discrete conversion using Zero-Order Hold (ZOH)
sys_d = c2d(sys_c, h, 'zoh');


Ad = sys_d.A;
Bd = sys_d.B;
Cd = sys_d.C;
Dd = sys_d.D;

% C and D stay the same
```

Discretized System Matrix Ad:

```
disp(Ad);
```

```
    1.0000   -0.0000   -0.0001    0.0100   -0.0000   -0.0000
         0    1.0011   -0.0009         0    0.0100   -0.0000
         0   -0.0007    1.0011         0   -0.0000    0.0100
         0   -0.0053   -0.0113    1.0000   -0.0000   -0.0001
         0    0.2252   -0.1782         0    1.0011   -0.0009
         0   -0.1400    0.2179         0   -0.0007    1.0011
```

Discretized Input Matrix Bd:

```
disp(Bd);
```

```
    0.0000   -0.0000   -0.0000
   -0.0000    0.0004   -0.0002
   -0.0000   -0.0002    0.0003
    0.0043   -0.0017   -0.0029
   -0.0017    0.0735   -0.0457
   -0.0029   -0.0457    0.0558
```

Sampling Period (Ts):

```
disp(h);
```

```
    0.0100
```

Problem 2.

(7 pts) First design an MPC using the augmented model without any constraints and implement the controller on the nonlinear continuous model. Write a script that animates the behavior of the closed-loop system.

The Augmented Model:

```
%MPC horizons
p = 100;
m = 10;

% Create the MPC object
mpc_obj = mpc(sys_d, h, p, m);
```

```
-->"Weights.ManipulatedVariables" is empty. Assuming default 0.00000.
-->"Weights.ManipulatedVariablesRate" is empty. Assuming default 0.10000.
-->"Weights.OutputVariables" is empty. Assuming default 1.00000.
```

```
% Weights: [x, theta1, theta2]
mpc_obj.Weights.OutputVariables = [2, 20, 20];
% Weights: [u1, u2, u3]
mpc_obj.Weights.ManipulatedVariables = [0.1, 0.1, 0.1];
% no constrains
```

```
mpc_obj.MV = struct('Min', -Inf, 'Max', +Inf);
```

```
-->The number of elements in the "ManipulatedVariables" property is smaller than 3. Assuming default value
```

```
mpc_obj.OV = struct('Min', -Inf, 'Max', +Inf);
```

```
-->The number of elements in the "OutputVariables" property is smaller than 3. Assuming default values.
```

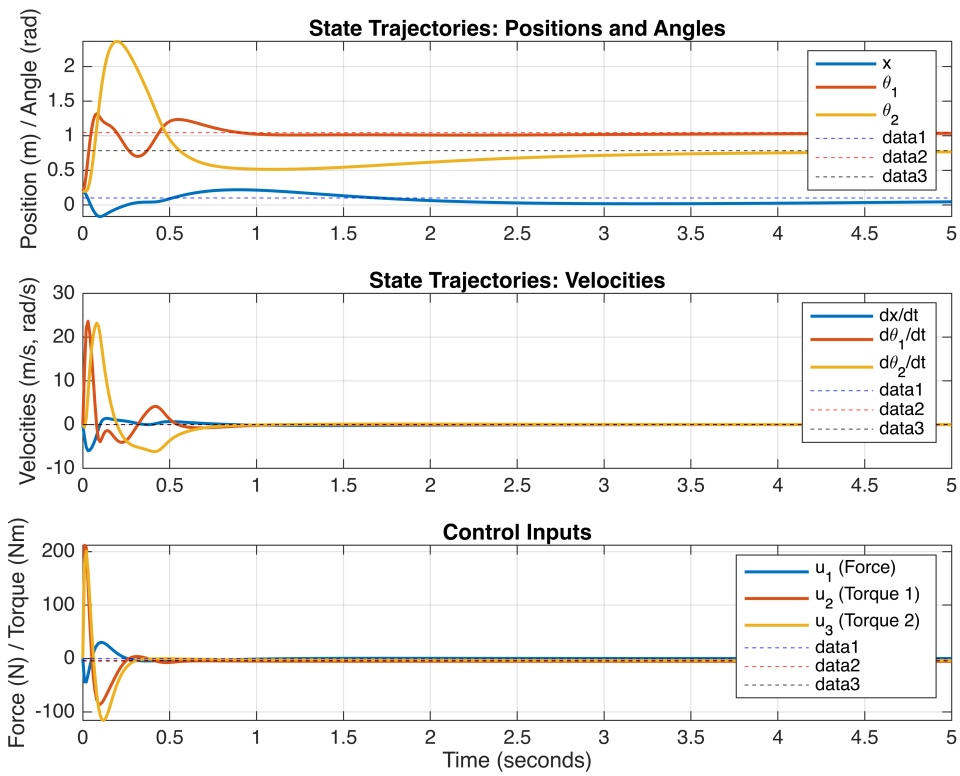
```
%equilibrium!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
mpc_obj.Model.Nominal.X = xe_vals;
mpc_obj.Model.Nominal.U = ue;
mpc_obj.Model.Nominal.Y = C_u3 * xe_vals;

x_init = [0.2; 0.2; 0.2; 0; 0; 0];
sim_time = 5;
r_target = C_u3 * xe_vals;

f_handle_3u = matlabFunction(f_u3, 'Vars', {state_vector, input_vector});
[T1, X1, U1] = run_mpc_simulation(mpc_obj, f_handle_3u, C_u3, x_init, h,
sim_time, r_target, xe_vals, ue, 'mpc_sim_P2.mp4');
```

```
Starting non-linear simulation...
-->Assuming output disturbance added to measured output #2 is integrated white noise.
-->Assuming output disturbance added to measured output #3 is integrated white noise.
   Assuming no disturbance added to measured output #1.
-->"Model.Noise" is empty. Assuming white noise on each measured output.
Simulation complete. Starting animation...
Animation video file saved as: mpc_sim_P2.mp4
```

Simulation Results

```
figure('Name', 'MPC Simulation Results without constrains');

subplot(3, 1, 1);
plot(T1, X1(:, 1:3), 'LineWidth', 1.5);
title('State Trajectories: Positions and Angles');
ylabel('Position (m) / Angle (rad)');
legend('x', '\theta_1', '\theta_2', 'Location', 'northeast');
yline(xe_vals(1), 'b--');
yline(xe_vals(2), 'r--');
yline(xe_vals(3), 'k--');
grid on;

subplot(3, 1, 2);
plot(T1, X1(:, 4:6), 'LineWidth', 1.5);
title('State Trajectories: Velocities');
ylabel('Velocities (m/s, rad/s)');
legend('dx/dt', 'd\theta_1/dt', 'd\theta_2/dt', 'Location', 'northeast');
yline(xe_vals(4), 'b--');
yline(xe_vals(5), 'r--');
yline(xe_vals(6), 'k--');
grid on;
```

```
subplot(3, 1, 3);
plot(T1, U1, 'LineWidth', 1.5);
title('Control Inputs');
ylabel('Force (N) / Torque (Nm)');
xlabel('Time (seconds)');
legend('u_1 (Force)', 'u_2 (Torque 1)', 'u_3 (Torque 2)', 'Location',
'northeast');
yline(ue(1), 'b--');
yline(ue(2), 'r--');
yline(ue(3), 'k--');
grid on;
```



Problem 3.

(7 pts) Impose constraints on the inputs and outputs. Re-design your MPC. Implement the controller on the nonlinear continuous model. Write a script that animates the behavior of the closed-loop system.

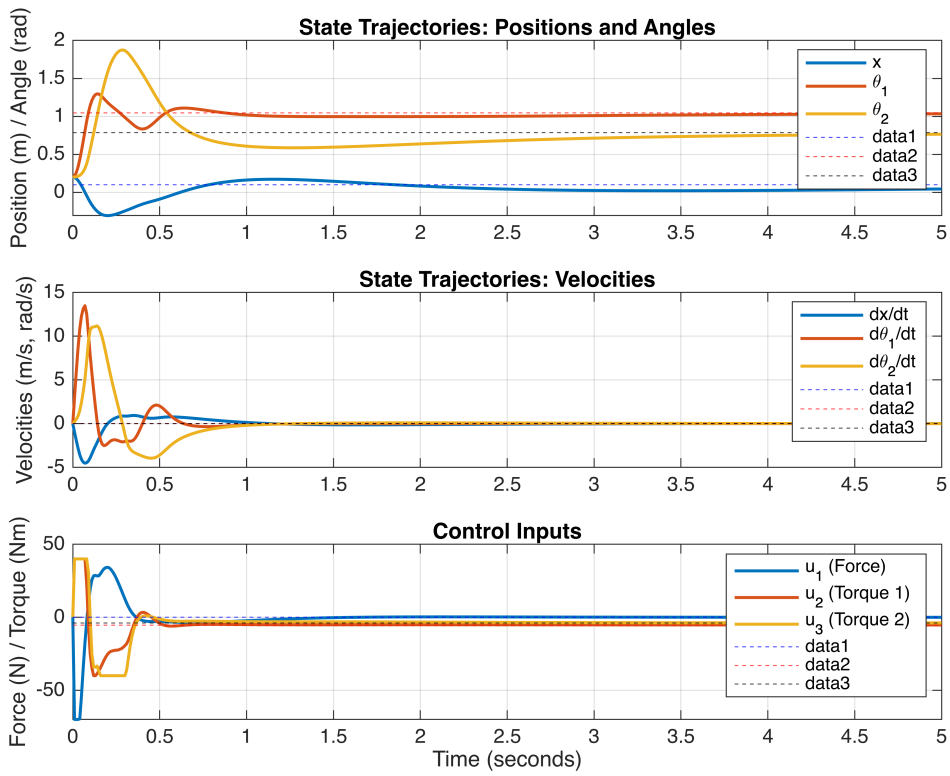
```
%Add constrains to model
mpc_obj.MV(1).Min = -70; mpc_obj.MV(1).Max = 70;
mpc_obj.MV(2).Min = -40; mpc_obj.MV(2).Max = 40;
mpc_obj.MV(3).Min = -40; mpc_obj.MV(3).Max = 40;


mpc_obj.OV(1).Min = -2; mpc_obj.OV(1).Max = 2;
```

```
mpc_obj.OV(2).Min = -2; mpc_obj.OV(2).Max = 2;
mpc_obj.OV(3).Min = -2; mpc_obj.OV(3).Max = 2;

f_handle_3u = matlabFunction(f_u3, 'Vars', {state_vector, input_vector});
[T1, X1, U1] = run_mpc_simulation(mpc_obj, f_handle_3u, C_u3, x_init, h,
sim_time, r_target, xe_vals, ue, 'mpc_sim_P3.mp4');
```

```
Starting non-linear simulation...
-->Assuming output disturbance added to measured output #2 is integrated white noise.
-->Assuming output disturbance added to measured output #3 is integrated white noise.
   Assuming no disturbance added to measured output #1.
-->"Model.Noise" is empty. Assuming white noise on each measured output.
Simulation complete. Starting animation...
Animation video file saved as: mpc_sim_P3.mp4
```

Simulation Results

```
figure('Name', 'MPC Simulation Results with constrains');

subplot(3, 1, 1);
plot(T1, X1(:, 1:3), 'LineWidth', 1.5);
title('State Trajectories: Positions and Angles');
ylabel('Position (m) / Angle (rad)');
legend('x', '\theta_1', '\theta_2', 'Location', 'northeast');
yline(xe_vals(1), 'b--');
yline(xe_vals(2), 'r--');
yline(xe_vals(3), 'k--');
grid on;

subplot(3, 1, 2);
plot(T1, X1(:, 4:6), 'LineWidth', 1.5);
title('State Trajectories: Velocities');
ylabel('Velocities (m/s, rad/s)');
legend('dx/dt', 'd\theta_1/dt', 'd\theta_2/dt', 'Location', 'northeast');
yline(xe_vals(4), 'b--');
yline(xe_vals(5), 'r--');

yline(xe_vals(6), 'k--');
grid on;

subplot(3, 1, 3);
plot(T1, U1, 'LineWidth', 1.5);
title('Control Inputs');
ylabel('Force (N) / Torque (Nm)');
xlabel('Time (seconds)');
legend('u_1 (Force)', 'u_2 (Torque 1)', 'u_3 (Torque 2)', 'Location',
'northeast');
yline(ue(1), 'b--');
yline(ue(2), 'r--');
yline(ue(3), 'k--');
grid on;
```



Problem 4.

(7 pts) Implement the combined MPC controller-observer compensator and test it on the nonlinear continuous-time (CT) model. Write a script that animates the behavior of the closed-loop system.

```
%using Oberserver from Funwork3

poles_obs_d = exp(eig(A_u3 - L*C_u3) * h); % Map continuous poles to
discrete poles
Ld = place(Ad', C_u3', poles_obs_d)';

disp(Ld)
```

```
    0.2453   -0.1049   -0.0153
    0.0879    0.2695   -0.0991
   -0.0284    0.1119    0.2417
    1.6081   -1.1410   -0.3221
    1.0032    2.2648   -1.3279
   -0.4275    1.0954    1.7555
```

```
eig(Ad-Ld*Cd)
```

```
ans = 6×1 complex
   0.8495 + 0.1573i
   0.8495 - 0.1573i
```

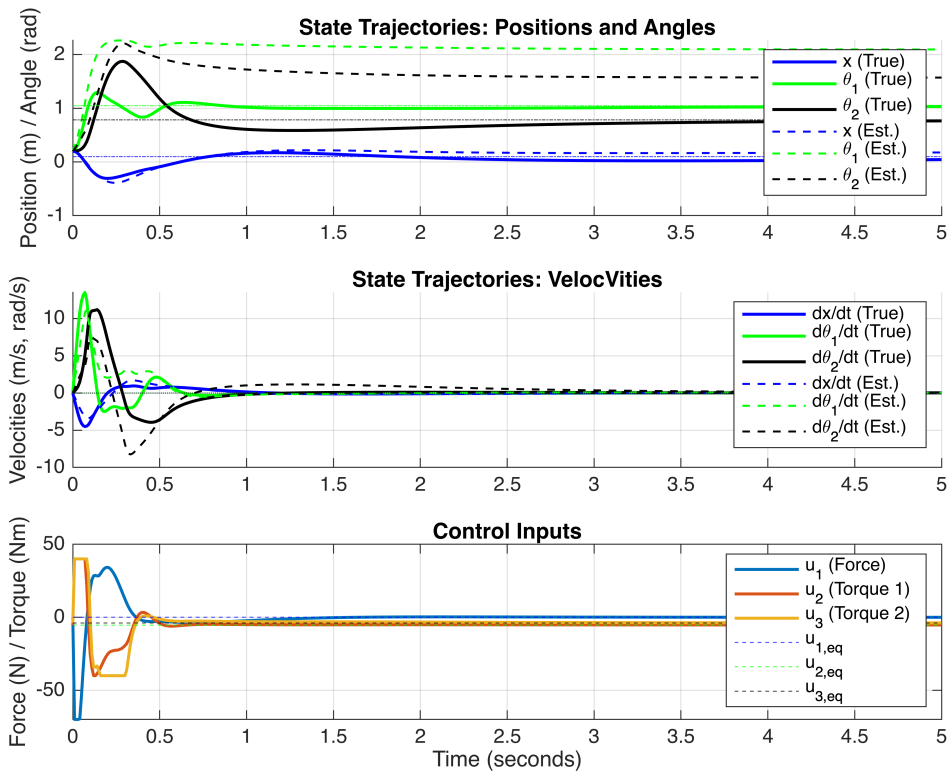
```
   0.9049 + 0.0000i
   0.8744 + 0.0000i
   0.8848 + 0.0151i
   0.8848 - 0.0151i
```

```
%using internal Observer --> Kalman filter
[T_obs, X_obs, U_obs, X_hat_obs] =
run_mpc_sim_with_Observer(mpc_obj,f_handle_3u, Ad, Bd, Cd, C_u3, x_init ,
Ld, h, sim_time, r_target, xe_vals, ue, 'mpc_sim_P4.mp4');
```

```
Starting DISCRETE plant simulation with MPC...
-->Assuming output disturbance added to measured output #2 is integrated white noise.
-->Assuming output disturbance added to measured output #3 is integrated white noise.
   Assuming no disturbance added to measured output #1.
-->"Model.Noise" is empty. Assuming white noise on each measured output.
Simulation complete. Starting animation...
Animation video file saved as: mpc_sim_P4.mp4
```

```
%External addtional Observer
%[T_obs, X_obs, U_obs, X_hat_obs] = run_mpc_with_Ld_observer(mpc_obj,
f_handle_3u, Ad, Bd, Cd, Ld, C_u3, x_init, h, sim_time, r_target, xe_vals,
ue, 'mpc_sim_P4.mp4')

figure('Name', 'MPC Discrete Simulation: True vs. Estimated States');

% Plot 1: Positions and Angles
subplot(3, 1, 1);
hold on;
% Plot True States (solid lines)
plot(T_obs, X_obs(:, 1), 'b-', 'LineWidth', 1.5);
plot(T_obs, X_obs(:, 2), 'g-', 'LineWidth', 1.5);
plot(T_obs, X_obs(:, 3), 'k-', 'LineWidth', 1.5);
% Plot Estimated States (dashed lines)
plot(T_obs, X_hat_obs(:, 1), 'b--', 'LineWidth', 1);
plot(T_obs, X_hat_obs(:, 2), 'g--', 'LineWidth', 1);
plot(T_obs, X_hat_obs(:, 3), 'k--', 'LineWidth', 1);
% Add equilibrium lines
yline(xe_vals(1), 'b-.');
yline(xe_vals(2), 'g-.');
yline(xe_vals(3), 'k-.');
hold off;
title('State Trajectories: Positions and Angles');
ylabel('Position (m) / Angle (rad)');
legend('x (True)', '\theta_1 (True)', '\theta_2 (True)', 'x (Est.)',
'\theta_1 (Est.)', '\theta_2 (Est.)', 'Location', 'northeast');
grid on;

% Plot 2: Velocities
subplot(3, 1, 2);
hold on;
% Plot True States (solid lines)
plot(T_obs, X_obs(:, 4), 'b-', 'LineWidth', 1.5);
```

```
plot(T_obs, X_obs(:, 5), 'g-', 'LineWidth', 1.5);
plot(T_obs, X_obs(:, 6), 'k-', 'LineWidth', 1.5);
% Plot Estimated States (dashed lines)
plot(T_obs, X_hat_obs(:, 4), 'b--', 'LineWidth', 1);
plot(T_obs, X_hat_obs(:, 5), 'g--', 'LineWidth', 1);
plot(T_obs, X_hat_obs(:, 6), 'k--', 'LineWidth', 1);
% Add equilibrium lines
yline(xe_vals(4), 'b:');
yline(xe_vals(5), 'g:');
yline(xe_vals(6), 'k:');
hold off;
title('State Trajectories: VelocVities');
ylabel('Velocities (m/s, rad/s)');
legend('dx/dt (True)', 'd\theta_1/dt (True)', 'd\theta_2/dt (True)',
'dx/dt (Est.)', 'd\theta_1/dt (Est.)', 'd\theta_2/dt (Est.)', 'Location',
'northeast');
grid on;

% Plot 3: Control Inputs
subplot(3, 1, 3);
plot(T_obs, U_obs, 'LineWidth', 1.5);
hold on;
% Add equilibrium input lines
yline(ue(1), 'b--');
yline(ue(2), 'g--');
yline(ue(3), 'k--');
hold off;
title('Control Inputs');
ylabel('Force (N) / Torque (Nm)');
xlabel('Time (seconds)');
legend('u_1 (Force)', 'u_2 (Torque 1)', 'u_3 (Torque 2)', 'u_{1,eq}',
'u_{2,eq}', 'u_{3,eq}', 'Location', 'northeast');
grid on;
```



Problem 4, Part 2.

(3 pts) Produce a video of your animation and submit your video to the FunWork 4 - Problem 4, Part 2 assignment link. (NOTE: This is the only part of your assignment that you do *not* need to submit to Gradescope.

Output for 3D Simulation in Simulink

```
%  physical parameters
l1 = 0.5;
l2 = 0.75;
g = 9.81;

% Cart dimensions
cart_width = 0.5;
cart_height = 0.25;

% the i use it with Fromworkspace blocks
x_res = timeseries(X_obs(:,1), T_obs);
Theta1_res = timeseries(X_obs(:,2), T_obs);
Theta2_res = timeseries(X_obs(:,3), T_obs);
```

Problem 5.

(12 pts) Repeat your MPC designs using the non-augmented model.

WITHOUT MATLAB TOOLBOX

started from sratch with Lecture notes becasue no differnce with matlab toolbox between Aug and NON Aug Model

```
n = size(Ad, 1); % n = 6
m = size(Bd, 2); % m = 3
p = size(Cd, 1); % p = 3



p_horizon = 100; % Prediction horizon (Np)

% Define cost matrices Q and R
Q0 = diag([2, 20, 20]);
R0 = 0.1 * eye(m);

% Create the large block-diagonal Q and R matrices for the horizon
Q_mpc = kron(eye(p_horizon), Q0);
R_mpc = kron(eye(p_horizon), R0);

% --- Calculate the W and Z matrices (Eq. 9, Page 3) ---
W = zeros(p_horizon * p, n);
Z = zeros(p_horizon * p, p_horizon * m);
temp_Z = zeros(p, p_horizon * m);

for i = 1:p_horizon
    % W = [C*A; C*A^2; ... C*A^p]
    W( (i-1)*p + 1 : i*p , : ) = Cd * (Ad^i);

    % Z = [CB, 0, ...; CAB, CB, ...; CA^pB, ..., CB]
    temp_Z = [Cd*(Ad^(i-1))*Bd, temp_Z(:, 1:end-m)];
    Z( (i-1)*p + 1 : i*p , : ) = temp_Z;
end


% K_inv = (R + Z'*Q*Z)^-1
K_inv = (R_mpc + Z' * Q_mpc * Z) \ eye(size(R_mpc));
% Kr = [I 0...0] * (R+Z'QZ)^-1 * Z'*Q
K_r = [eye(m), zeros(m, (p_horizon-1)*m)] * K_inv * Z' * Q_mpc
```

```
K_r = 3×300
```

```
    0.0004   -0.0017   -0.0026    0.0012   -0.0049   -0.0076    0.0021   -0.0080 ...
   -0.0002    0.0610   -0.0365   -0.0005    0.1719   -0.1015   -0.0008    0.2619
   -0.0003   -0.0364    0.0467   -0.0007   -0.1010    0.1321   -0.0012   -0.1504
```

```
% Kx = Kr * W
K_x = K_r * W
```

```
K_x = 3×6
    2.9967   -1.4979   -1.9783    1.9608   -0.2274   -0.2766
    0.2028   16.3890    0.0537    0.1991    2.7273    1.2454
    0.2753    0.4469   17.4495    0.2667    1.3097    3.3155
```

USE the Discrete-Time Observer

```
disp(Ld)
```

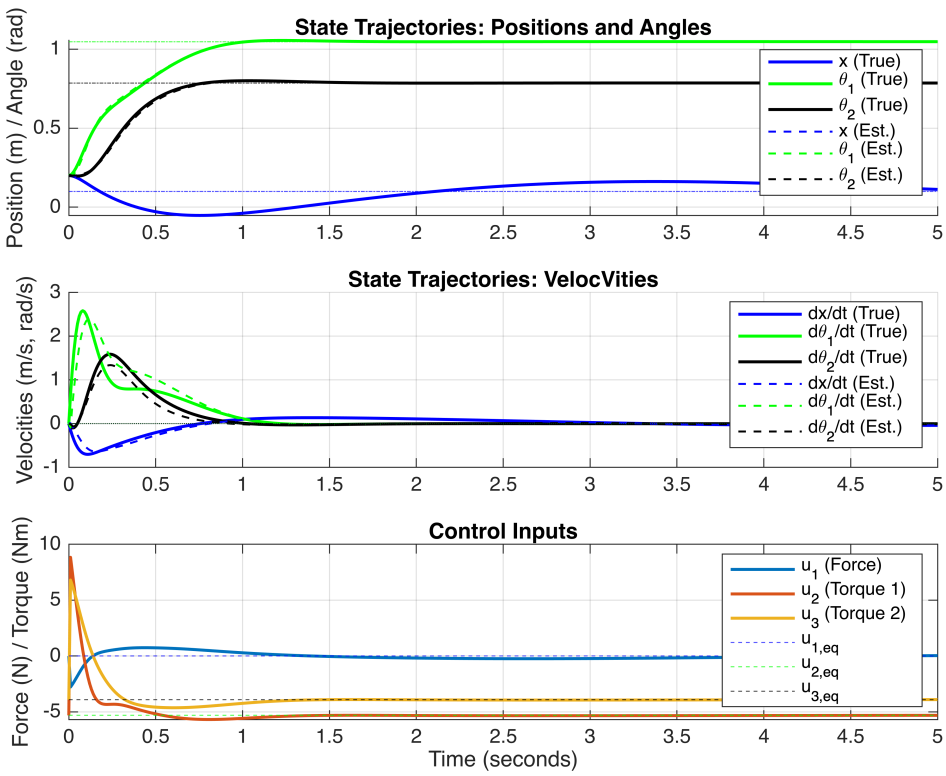
```
    0.2453   -0.1049   -0.0153
    0.0879    0.2695   -0.0991
   -0.0284    0.1119    0.2417
    1.6081   -1.1410   -0.3221
    1.0032    2.2648   -1.3279
   -0.4275    1.0954    1.7555
```

```
%Simulation
[T_obs_NA, X_obs_NA, X_hat_obs_NA, U_obs_NA] =
run_manual_mpc_sim(f_handle_3u, Ad, Bd, Cd, K_x, K_r, Ld, x_init, h,
sim_time, r_target, xe_vals, ue, 'manual_mpc_sim_NONAUG.mp4');
```

```
Starting manual MPC + Observer simulation...
Simulation complete. Starting animation...
Animation video file saved as: manual_mpc_sim_NONAUG.mp4
```

```
figure('Name', 'NonAugmMPC Simulation: True vs. Estimated States');

% Plot 1: Positions and Angles
subplot(3, 1, 1);
hold on;
% Plot True States (solid lines)
plot(T_obs_NA, X_obs_NA(:, 1), 'b-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 2), 'g-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 3), 'k-', 'LineWidth', 1.5);
% Plot Estimated States (dashed lines)
plot(T_obs_NA, X_hat_obs_NA(:, 1), 'b--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 2), 'g--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 3), 'k--', 'LineWidth', 1);
% Add equilibrium lines
yline(xe_vals(1), 'b-.');
yline(xe_vals(2), 'g-.');
yline(xe_vals(3), 'k-.');
hold off;
title('State Trajectories: Positions and Angles');
ylabel('Position (m) / Angle (rad)');
legend('x (True)', '\theta_1 (True)', '\theta_2 (True)', 'x (Est.)',
'\theta_1 (Est.)', '\theta_2 (Est.)', 'Location', 'northeast');
```

```
grid on;

% Plot 2: Velocities
subplot(3, 1, 2);
hold on;
% Plot True States (solid lines)
plot(T_obs_NA, X_obs_NA(:, 4), 'b-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 5), 'g-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 6), 'k-', 'LineWidth', 1.5);
% Plot Estimated States (dashed lines)
plot(T_obs_NA, X_hat_obs_NA(:, 4), 'b--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 5), 'g--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 6), 'k--', 'LineWidth', 1);
% Add equilibrium lines
yline(xe_vals(4), 'b:');
yline(xe_vals(5), 'g:');
yline(xe_vals(6), 'k:');
hold off;
title('State Trajectories: VelocVities');
ylabel('Velocities (m/s, rad/s)');
legend('dx/dt (True)', 'd\theta_1/dt (True)', 'd\theta_2/dt (True)',
'dx/dt (Est.)', 'd\theta_1/dt (Est.)', 'd\theta_2/dt (Est.)', 'Location',
'northeast');
grid on;

% Plot 3: Control Inputs
subplot(3, 1, 3);
plot(T_obs_NA, U_obs_NA, 'LineWidth', 1.5);
hold on;
% Add equilibrium input lines
yline(ue(1), 'b--');
yline(ue(2), 'g--');
yline(ue(3), 'k--');
hold off;
title('Control Inputs');
ylabel('Force (N) / Torque (Nm)');
xlabel('Time (seconds)');
legend('u_1 (Force)', 'u_2 (Torque 1)', 'u_3 (Torque 2)', 'u_{1,eq}',
'u_{2,eq}', 'u_{3,eq}', 'Location', 'northeast');
grid on;
```




```
return
```

WITH MATLAB TOOLBOX

```
% integartor behaviour deaktivated
mpc_obj.Model.Disturbance = [];

[T_obs_NA, X_obs_NA, U_obs_NA, X_hat_obs_NA] =
run_mpc_sim_with_Observer(mpc_obj,f_handle_3u, Ad, Bd, Cd, C_u3, x_init ,
h, sim_time, r_target, xe_vals, ue, 'mpc_sim_P4.mp4');

figure('Name', 'MPC Discrete Simulation: True vs. Estimated States');

% Plot 1: Positions and Angles
subplot(3, 1, 1);
hold on;
% Plot True States (solid lines)
plot(T_obs_NA, X_obs_NA(:, 1), 'b-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 2), 'g-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 3), 'k-', 'LineWidth', 1.5);
% Plot Estimated States (dashed lines)
plot(T_obs_NA, X_hat_obs_NA(:, 1), 'b--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 2), 'g--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 3), 'k--', 'LineWidth', 1);
% Add equilibrium lines
```

```
yline(xe_vals(1), 'b-.');
yline(xe_vals(2), 'g-.');
yline(xe_vals(3), 'k-.');
hold off;
title('State Trajectories: Positions and Angles');
ylabel('Position (m) / Angle (rad)');
legend('x (True)', '\theta_1 (True)', '\theta_2 (True)', 'x (Est.)',
'\theta_1 (Est.)', '\theta_2 (Est.)', 'Location', 'northeast');
grid on;

% Plot 2: Velocities
subplot(3, 1, 2);
hold on;
% Plot True States (solid lines)
plot(T_obs_NA, X_obs_NA(:, 4), 'b-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 5), 'g-', 'LineWidth', 1.5);
plot(T_obs_NA, X_obs_NA(:, 6), 'k-', 'LineWidth', 1.5);
% Plot Estimated States (dashed lines)
plot(T_obs_NA, X_hat_obs_NA(:, 4), 'b--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 5), 'g--', 'LineWidth', 1);
plot(T_obs_NA, X_hat_obs_NA(:, 6), 'k--', 'LineWidth', 1);
% Add equilibrium lines
yline(xe_vals(4), 'b:');
yline(xe_vals(5), 'g:');
yline(xe_vals(6), 'k:');
hold off;
title('State Trajectories: VelocVities');
ylabel('Velocities (m/s, rad/s)');
legend('dx/dt (True)', 'd\theta_1/dt (True)', 'd\theta_2/dt (True)',
'dx/dt (Est.)', 'd\theta_1/dt (Est.)', 'd\theta_2/dt (Est.)', 'Location',
'northeast');
grid on;

% Plot 3: Control Inputs
subplot(3, 1, 3);
plot(T_obs_NA, U_obs_NA, 'LineWidth', 1.5);
hold on;
% Add equilibrium input lines
yline(ue(1), 'b--');
yline(ue(2), 'g--');
yline(ue(3), 'k--');
hold off;
title('Control Inputs');
ylabel('Force (N) / Torque (Nm)');
xlabel('Time (seconds)');
legend('u_1 (Force)', 'u_2 (Torque 1)', 'u_3 (Torque 2)', 'u_{1,eq}',
'u_{2,eq}', 'u_{3,eq}', 'Location', 'northeast');
grid on;
```